

Rydberg gas as an RF antenna

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This abstract is abstract.

I. INTRODUCTION

II. THEORY

A. Hamiltonian description

The four eigenstates of the bare atomic Hamiltonian H_0 involved in the DEIT process are $\{|1\rangle, |2\rangle, |3\rangle, |4\rangle\}$, with associated energies $\hbar\omega_j$, i.e. $H_0|j\rangle = \hbar\omega_j|j\rangle$. Including the interaction provoked by the external electric fields $\mathbf{E}_i$, the full Hamiltonian in the lab frame takes the form

$$H = H_0 + H_E, \quad H_E(t) \equiv -\hat{\mathbf{d}} \cdot \sum_i \mathbf{E}_i(t),$$

where $\hat{\mathbf{d}}$ is the electric dipole moment of the Rydberg atom. The atom is exposed to three different electric fields; a weak probe field $\mathbf{E}_p$ is established by the probe laser, an intense dressing field $\mathbf{E}_c$ is generated by the coupling laser, and a separate field $\mathbf{E}_{\text{RF}}$ is created by the RF source.

For simplicity, we treat all fields f as plane waves,

$$\mathbf{E}_f = \frac{\boldsymbol{\varepsilon}_f}{2} (e^{i\omega_f t} + e^{-i\omega_f t}).$$

The probe laser drives the $|1\rangle \rightarrow |2\rangle$ transition, the coupling laser drives $|2\rangle \rightarrow |3\rangle$, and the RF pumping field drives $|3\rangle \rightarrow |4\rangle$. Accordingly, the field frequencies are near resonance with their respective transitions: $\omega_p \simeq \omega_{21}$, $\omega_c \simeq \omega_{32}$, and $\omega_{\text{RF}} \simeq \omega_{43}$, where we define $\omega_{nm} = \omega_n - \omega_m$ to be the transition frequency from $|m\rangle$ to $|n\rangle$.

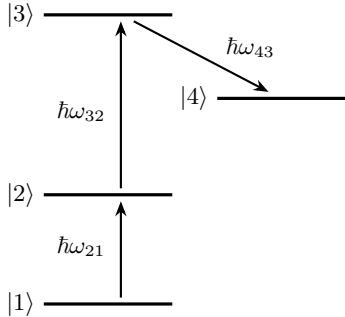


Figure 1. Bare atom energy levels... work in progress...

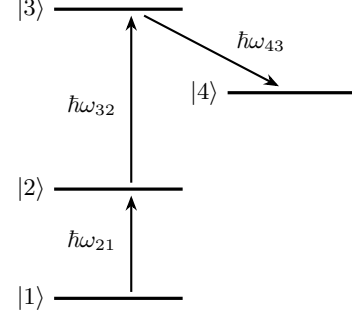


Figure 2. Dressed states... work in progress...

We will also need the Rabi frequencies associated with each transition. They are in our case given by

$$\Omega_p = \frac{\langle 2 | \hat{\mathbf{d}} \cdot \boldsymbol{\varepsilon}_p | 1 \rangle}{\hbar} = \frac{\wp_p |E_p|}{\hbar},$$

$$\Omega_c = \frac{\langle 3 | \hat{\mathbf{d}} \cdot \boldsymbol{\varepsilon}_c | 2 \rangle}{\hbar} = \frac{\wp_c |E_c|}{\hbar},$$

$$\Omega_{\text{RF}} = \frac{\langle 4 | \hat{\mathbf{d}} \cdot \boldsymbol{\varepsilon}_{\text{RF}} | 3 \rangle}{\hbar} = \frac{\wp_{\text{RF}} |E_{\text{RF}}|}{\hbar}.$$

Here we follow Holloway et al. [1], defining

$$\wp_f = \langle j | \hat{\mathbf{d}} | i \rangle = e\epsilon \cdot \int \psi_i^* \mathbf{r} \psi_j d^3r,$$

where e is the elemental charge, ϵ is polarisation vector of the laser f , and ψ_i, ψ_j ($i < j$) are the wavefunctions of the states coupled by the field f .

Working in the interaction picture with respect to the bare Hamiltonian H_0 , a general state of the atom in this setup is given by

$$|\psi(t)\rangle = \sum_{n=1}^4 c_n e^{-iH_0 t/\hbar} |n\rangle = \sum_{n=1}^4 c_n e^{-i\omega_n t} |n\rangle,$$

$$\partial_t |\psi(t)\rangle = \sum_{n=1}^4 (\dot{c}_n - i\omega_n c_n) e^{-i\omega_n t} |n\rangle.$$

The Schrödinger equation then reads $i\hbar\partial_t |\psi\rangle = \hat{H}_E |\psi\rangle$, where $\hat{H}_E$ is the interaction Hamiltonian in the interaction picture, i.e.

$$\hat{H}_E(t) = e^{iH_0 t/\hbar} H_E(t) e^{-iH_0 t/\hbar}.$$

The matrix elements of the interaction picture Hamiltonian in the $\{|1\rangle, |2\rangle, |3\rangle, |4\rangle\}$ basis are given by

$$(\hat{H}_E)_{nm} = \langle n | \hat{H}_E | m \rangle = e^{i\omega_{nm} t} \langle n | H_E | m \rangle.$$

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The non-zero elements are those associated with the three driven transitions. As an example we can calculate the (1,2)-element:

$$\begin{aligned} (\hat{H}_E)_{12} &= e^{i\omega_{12}t} \langle 1 | H_E | 2 \rangle \\ &= \frac{1}{2} e^{-i\omega_{21}t} \langle 1 | \hat{\mathbf{d}} \cdot \boldsymbol{\mathcal{E}}_p | 2 \rangle (e^{i\omega_p t} + e^{-i\omega_p t}) \\ &= \frac{1}{2} \hbar \Omega_p (e^{-i(\omega_{21}-\omega_p)t} + e^{-i(\omega_{21}+\omega_p)t}). \end{aligned}$$

The rapidly oscillating $e^{-i(\omega_{21}+\omega_p)t}$ term will be washed out and averaged to zero over any reasonable measurement period and can thus be neglected. Rewriting the surviving term using the detuning parameter $\Delta_p = \omega_{21} - \omega_p$, we find

$$(\hat{H}_E)_{12} = (\hat{H}_E)_{12}^\dagger = \frac{1}{2} \hbar \Omega_p e^{-i\Delta_p t}.$$

The other matrix elements can be found completely analogously and in the end we get

$$\hat{H}_E = \frac{\hbar}{2} \begin{bmatrix} 0 & \Omega_p e^{-i\Delta_p t} & 0 & 0 \\ \Omega_p e^{i\Delta_p t} & 0 & \Omega_c e^{-i\Delta_c t} & 0 \\ 0 & \Omega_c e^{i\Delta_c t} & 0 & \Omega_{\text{RF}} e^{-i\Delta_{\text{RF}} t} \\ 0 & 0 & \Omega_{\text{RF}} e^{i\Delta_{\text{RF}} t} & 0 \end{bmatrix}.$$

We can get rid of the time-dependence of the Hamiltonian by transforming to a rotating reference frame using the unitary transformation matrix

$$U \equiv \text{diag}(1, e^{-i\Delta_p t}, e^{-i(\Delta_p + \Delta_c)t}, e^{-i(\Delta_p + \Delta_c + \Delta_{\text{RF}})t}).$$

The resulting Hamiltonian in the rotating frame can be determined to be

$$\begin{aligned} \mathcal{H} &= U^\dagger \hat{H}_E U - i\hbar U^\dagger \dot{U} \\ &= \hbar \begin{bmatrix} 0 & \Omega_p/2 & 0 & 0 \\ \Omega_p/2 & \Delta_p & \Omega_c/2 & 0 \\ 0 & \Omega_c/2 & \Delta_p + \Delta_c & \Omega_{\text{RF}}/2 \\ 0 & 0 & \Omega_{\text{RF}}/2 & \Delta_p + \Delta_c + \Delta_{\text{RF}} \end{bmatrix}. \quad (1) \end{aligned}$$

1. Including decay: non-Hermitian Hamiltonian

TO BE WRITTEN...

$$H_{\text{dec}} = -\frac{i\hbar}{2} \sum_{n=2}^4 \gamma_n |n\rangle\langle n|$$

B. Eigenstates in the presence of the coupling field

By design, we keep the probe field weak which means that its Rabi frequency is much smaller than the coupling frequency, i.e. $\Omega_p \ll \Omega_c$. This in turn means that the dynamics on the $\{|2\rangle, |3\rangle, |4\rangle\}$ manifold evolve on much faster timescale than the probe-induced transition out

of the ground state $|1\rangle$. The ground state remains therefore nearly fully populated and effectively acts as a reservoir. One can thus, to leading order, treat the strong-field problem first, setting up so-called dressed states on the $\{|2\rangle, |3\rangle, |4\rangle\}$ manifold, and then introduce the probe field perturbatively as a weak coupling between the $|1\rangle$ state and the dressed states.

The Hermitian part of the strong-field Hamiltonian in the rotating frame is just the lower (3×3) -submatrix of Eq.(1). As a simplification for our analytical approach, we will consider the case with perfectly on-resonance coupling and RF pumping, i.e. $\Delta_c = \Delta_{\text{RF}} = 0$. This gives us the strong-field Hamiltonian

$$\mathcal{H}_s = \hbar \begin{bmatrix} \Delta_p & \Omega_c/2 & 0 \\ \Omega_c/2 & \Delta_p & \Omega_{\text{RF}}/2 \\ 0 & \Omega_{\text{RF}}/2 & \Delta_p \end{bmatrix}.$$

Since the Δ_p 's along the diagonal will uniformly shift the energy levels, the energies of the eigenstates of $\mathcal{H}_s$ will be on the form $\varepsilon_n = \hbar(\Delta_p + \lambda_n)$. We can thus solve for the λ 's by finding the eigenvalues of the strong-field Hamiltonian with $\Delta_p = 0$ along the diagonal. Doing this, we get three eigenstates with eigenvalues

$$\lambda_0 = 0, \quad \lambda_{\pm} = \pm \frac{\Omega_{\text{eff}}}{2},$$

with $\Omega_{\text{eff}} \equiv \sqrt{\Omega_{\text{RF}}^2 + \Omega_c^2}$. The eigenstate associated with λ_0 is the so-called *dark state* and can be represented as the dressed state

$$|\Psi_0\rangle = -\frac{\Omega_{\text{RF}}}{\Omega_{\text{eff}}} |2\rangle + \frac{\Omega_c}{\Omega_{\text{eff}}} |4\rangle.$$

The two other eigenstates, the *bright states*, can be represented as

$$|\Psi_{\pm}\rangle = \frac{\Omega_c}{\sqrt{2}\Omega_{\text{eff}}} |2\rangle \pm \frac{1}{\sqrt{2}} |3\rangle + \frac{\Omega_{\text{RF}}}{\sqrt{2}\Omega_{\text{eff}}} |4\rangle.$$

C. The susceptibility of the gas

CLARIFY NOTATION...

When we turn on the probe laser that couples the $|1\rangle$ state to $|2\rangle$, the gas becomes opaque due to the strong absorption at the probe frequency. However, when we turn on the coupling laser (and the RF signal), the $|2\rangle$ state is no longer an eigenstate of the system and the probe laser couples the ground state to the set of dressed states $\{|\Psi_n\rangle\}$ instead. This effectively makes the gas transparent as seen by the probe laser. Let us quantify this change in opacity due to the coupling field and the RF signal.

The susceptibility χ is the first order correction to the permittivity, $\epsilon_0 \rightarrow \epsilon_0(1 + \chi)$. When the probe field $\mathbf{E}_p$ is applied to the gas, it sets up a global polarisation

$$\mathbf{P}(t) = \epsilon_0 \chi \mathbf{E}_p(t). \quad (2)$$

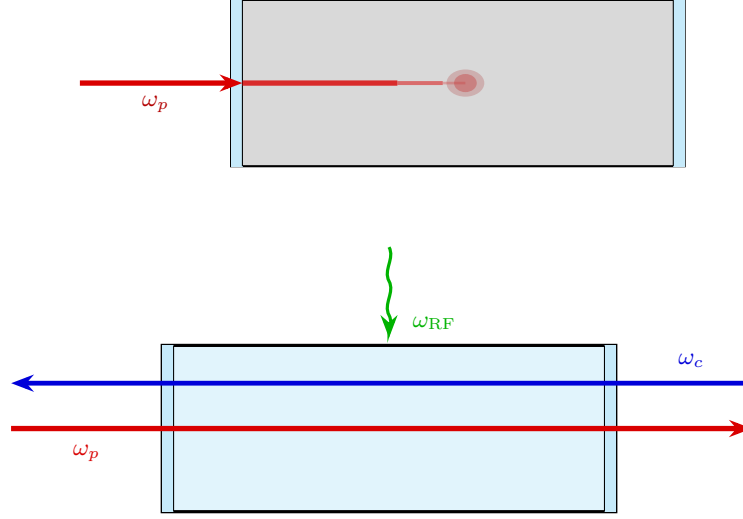


Figure 3. EIT... work in progress...

From a microscopic description, the same polarisation vector can be derived from the expectation value of the dipole moments of the Rydberg atoms in the cell

$$P(t) = N \langle \hat{d} \rangle_t = N \text{Tr} [\rho \hat{\mathbf{d}}] \Big|_t, \quad (3)$$

where N is the number density of the Rydberg atoms in the cell and ρ is the density matrix calculated to first order since we treat $\mathbf{E}_p$ as a weak field and therefore a first order perturbation.

Putting the macroscopic and microscopic formulations, Eqs.(2) and (3), of the polarisation, we find the susceptibility as

$$\chi = \frac{N}{\epsilon_0} \frac{\text{Tr} [\rho \hat{\mathbf{d}}] \Big|_t}{|E_p|}. \quad (4)$$

1. The density matrix from Lindblad's master equation

In the rotating frame, strong-field Hamiltonian is $\mathcal{H}_s = \mathcal{H}_0 + \mathcal{H}_E$. The ground state has zero energy in this frame, so

$$\mathcal{H}_0 = \sum_{n=0,+,-} \varepsilon_n |\Psi_n\rangle \langle \Psi_n|,$$

and the interaction part is

$$\mathcal{H}_E = \sum_{f=p,c,\text{RF}} \frac{\hbar \Omega_f}{2} |n\rangle \langle m| + \text{h.c.},$$

where $|n\rangle, |m\rangle$ are the two states coupled by the field f .

The density matrix can be found from Lindblad's master equation [2], taking the form as the ODE

$$\dot{\rho} = -\frac{i}{\hbar} [\mathcal{H}_s, \rho] + \mathcal{L}_{\text{dec}}[\rho],$$

with the decay term in the basis of the bare eigenstates

$$\mathcal{L}_{\text{dec}}[\rho] \equiv \sum_{n>m} \Gamma_{nm} \left(|m\rangle \langle n| \rho |n\rangle \langle m| - \frac{1}{2} \{ |n\rangle \langle n|, \rho \} \right).$$

After strong coupling is turned on, the probe sends the ground state $|1\rangle$ onto the manifold of dressed states $\{|\Psi_n\rangle\}$. Therefore, the only non-zero elements in the basis of dressed states are the coherences between the dressed states and the ground state,

$$\begin{aligned} \dot{\rho}_{n1} &= \langle \Psi_n | \dot{\rho} | 1 \rangle \\ &= -\frac{i}{\hbar} \langle \Psi_n | [\mathcal{H}_s, \rho] | 1 \rangle + \langle \Psi_n | \mathcal{L}_{\text{dec}} | 1 \rangle. \end{aligned}$$

In the first term, we have two possible contributions, the first of which comes from

$$\begin{aligned} \langle \Psi_n | \mathcal{H}_s \rho | 1 \rangle &= \langle \Psi_n | \mathcal{H}_s | 1 \rangle \langle 1 | \rho | 1 \rangle + \sum_m \langle \Psi_n | \mathcal{H}_s | \Psi_m \rangle \langle \Psi_m | \rho | 1 \rangle \\ &= \langle \Psi_n | \mathcal{H}_E | 1 \rangle \rho_{11} + \langle \Psi_n | \mathcal{H}_0 | \Psi_n \rangle \rho_{n1} \\ &= \frac{\hbar \Omega_p}{2} c_n^* + \varepsilon_n \rho_{n1}, \end{aligned}$$

where $c_n = \langle 2 | \Psi_n \rangle$ coefficients are the projections of the dressed states onto the $|2\rangle$ state. Here we have neglected the higher order corrections from $\langle \Psi_n | \mathcal{H}_E | \Psi_m \rangle$ and that the ground state is almost fully occupied, since the probe laser is weak, such that $\rho_{11} \simeq 1$ to first order.

The second possible contribution to the first term comes from $\langle \Psi_n | \rho \mathcal{H}_s | 1 \rangle$. However, this enters only at second order in the perturbations, so we neglect it.

The projection of the decay term onto the ground state $|1\rangle$ kills the first term in the sum and leaves the anticor-

mutator part as the only surviving terms. Thus

$$\begin{aligned}
\langle \Psi_n | \mathcal{L}_{\text{dec}} | 1 \rangle &= -\frac{1}{2} \sum_{i>j} \Gamma_{ij} \left(\langle \Psi_n | i \rangle \langle i | \rho | 1 \rangle + \underbrace{\langle \Psi_n | \rho | i \rangle \langle i | 1 \rangle}_0 \right) \\
&= -\frac{1}{2} \sum_{i>j} \Gamma_{ij} \langle \Psi_n | i \rangle \sum_m \langle i | \Psi_m \rangle \langle \Psi_m | \rho | 1 \rangle \\
&= -\frac{1}{2} \sum_m \left(\sum_{i>j} \Gamma_{ij} \langle \Psi_n | i \rangle \langle i | \Psi_m \rangle \right) \rho_{m1}.
\end{aligned}$$

Assuming that the decay rates Γ_{nm} (the big parenthesis) are small relative to the energy splitting of the dressed states, we can apply the secular approximation $\Gamma_{nm} \simeq \gamma_n \delta_{nm}$, where $\gamma_n = \sum_m \Gamma_{nm}$. **JUSTIFY MORE AND CITE...** This gives the neat result

$$\langle \Psi_n | \mathcal{L}_{\text{dec}} | 1 \rangle = -\gamma_n \rho_{n1} / 2.$$

Collecting the terms we have calculated, the Lindblad equation reads

$$\dot{\rho}_{n1} = \left[i\varepsilon_n / \hbar - \frac{\gamma_n}{2} \right] \rho_{n1} - \frac{i\Omega_p}{2} c_n^*.$$

The homogeneous solution yields a rapidly decaying oscillating coherence $\rho_{n1}^{\text{hom.}}(t) = \rho_{n1}(0) e^{-\gamma_n t/2} e^{i\varepsilon_n t/\hbar}$, which we can neglect for $t \gg 1/\gamma_n$. For the particular solution, we consider the steady-state solution $\dot{\rho}_{n1} = 0$. This gives our final expression for the density matrix elements

$$\rho_{n1}(t) = \frac{\Omega_p c_n^*}{2\varepsilon_n / \hbar - i\gamma_n}, \quad t \gg 1/\gamma_n.$$

2. Evaluating the susceptibility

Going back to the macroscopic description of the polarisation, as given in Eq.(2), we can write the a sum of two oscillating terms with opposite frequency

$$P(t) = \epsilon_0 \chi E_p(t) = \mathcal{P} e^{-i\omega_p t} + \text{c.c.},$$

where $\mathcal{P} = \epsilon_0 \chi E_p / 2$. In the microscopic description, we similarly separate the expression into two complex conjugate terms. This can be done in the trace term, since

$$\begin{aligned}
\text{Tr}[\rho \hat{\mathbf{d}}] &= \langle 1 | \rho \hat{\mathbf{d}} | 1 \rangle + \sum_m \langle \Psi_m | \rho \hat{\mathbf{d}} | \Psi_m \rangle \\
&= \sum_n \left(\langle 1 | \rho | \Psi_n \rangle \langle \Psi_n | \hat{\mathbf{d}} | 1 \rangle \right. \\
&\quad \left. + \langle \Psi_n | \rho | 1 \rangle \langle 1 | \hat{\mathbf{d}} | \Psi_n \rangle \right) \\
&= \sum_n \rho_{n1} \hat{d}_{1n} + \text{c.c.}
\end{aligned}$$

Therefore, we can use $\mathcal{P} = \epsilon_0 \chi E_p / 2 = N \sum_n \rho_{n1} \hat{d}_{1n}$ in order to get

$$\begin{aligned}
\chi &= \frac{2N}{\epsilon_0 E_p} \sum_n \rho_{n1} \hat{d}_{1n} \\
&= \frac{2N}{\epsilon_0 E_p} \sum_n \frac{\Omega_p c_n^* \hat{d}_{1n}}{2\varepsilon_n / \hbar - i\gamma_n} \\
\therefore \chi &= \frac{N \mathcal{P}_p^2}{\epsilon_0} \sum_n \frac{|c_n|^2}{\varepsilon_n - i\hbar\gamma_n/2}. \quad (5)
\end{aligned}$$

For the last equality we use a resolution of the identity to get $\hat{d}_{1n} = \langle 1 | \hat{\mathbf{d}} | 2 \rangle \langle 2 | \Psi_n \rangle = \hbar \Omega_p c_n / E_p$ and the definition of the Rabi frequency.

D. Appearance of two EIT windows

REWRITE SLIGHTLY...

The transition amplitudes of the probe to each of the dressed states is now given by the coupling constants and can be found as the projection of the $|2\rangle$ state onto each of the eigenstates,

$$\begin{aligned}
c_0 &= \langle 2 | \Psi_0 \rangle = -\frac{\Omega_{\text{RF}}}{\Omega_{\text{eff}}}, \\
c_{\pm} &= \langle 2 | \Psi_{\pm} \rangle = \frac{\Omega_c}{\sqrt{2}\Omega_{\text{eff}}}.
\end{aligned}$$

The susceptibility is shown to be

$$\chi \propto \sum_{n=0,-,+} \frac{|c_n|^2}{\Delta_p - \lambda_n},$$

which in our case can be expanded and simplified to

$$\chi \propto \frac{\Delta_p^2 - \Omega_{\text{RF}}^2/4}{\Delta_p(\Delta_p^2 - \Omega_{\text{eff}}^2/4)}.$$

The absorption peaks when the denominator goes to zero. We therefore have three such peaks, corresponding to $\Delta_p = 0$ and $\Delta_p = \pm\Omega_{\text{eff}}/2$. Crucially, we also get two transparency windows around $\chi = 0$ which are centered at $\Delta_p = \pm\Omega_{\text{RF}}/2$. These are the two EIT windows.

E. Doppler effect for warm atoms

III. METHOD

A. Simulating the gas using Rydqule

IV. RESULTS

V. DISCUSSION

VI. CONCLUSION

ACKNOWLEDGMENTS

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- [1] Christopher L. Holloway, Joshua A. Gordon, Steven Jefferts, Andrew Schwarzkopf, David A. Anderson, Stephanie A. Miller, Nithiwadee Thaicharoen, and Georg Raithel. Broadband rydberg atom-based electric-field probe for si-traceable, self-calibrated measurements. *IEEE Transactions on Antennas and Propagation*, 62(12):6169–6182, 2014.
- [2] Daniel Manzano. A short introduction to the lindblad master equation. *AIP Advances*, 10(2), February 2020.